

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456339.90215 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.014
Transit depth (Rp/Rs)^2	2.48 +/- 0.44 [%]
Orbital Inclination (inc)	82.99 +/- 1.6
Airmass coefficient 1 (a1)	0.946 +/- 0.012
Airmass coefficient 2 (a2)	0.0348 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	#1 - [401, 372]
Optimal Method	PSF photometry
Transit Duration (day)	0.0527 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.157 ± 0.014 vs 0.1594 (deviation 0.17σ)
Duration fit vs published	0.0527 d vs 0.0512 d (deviation 0.32σ)
Mid-time O-C	-0.3 min
Depth SNR	11.2
Residual scatter	1.27 %

Light curve

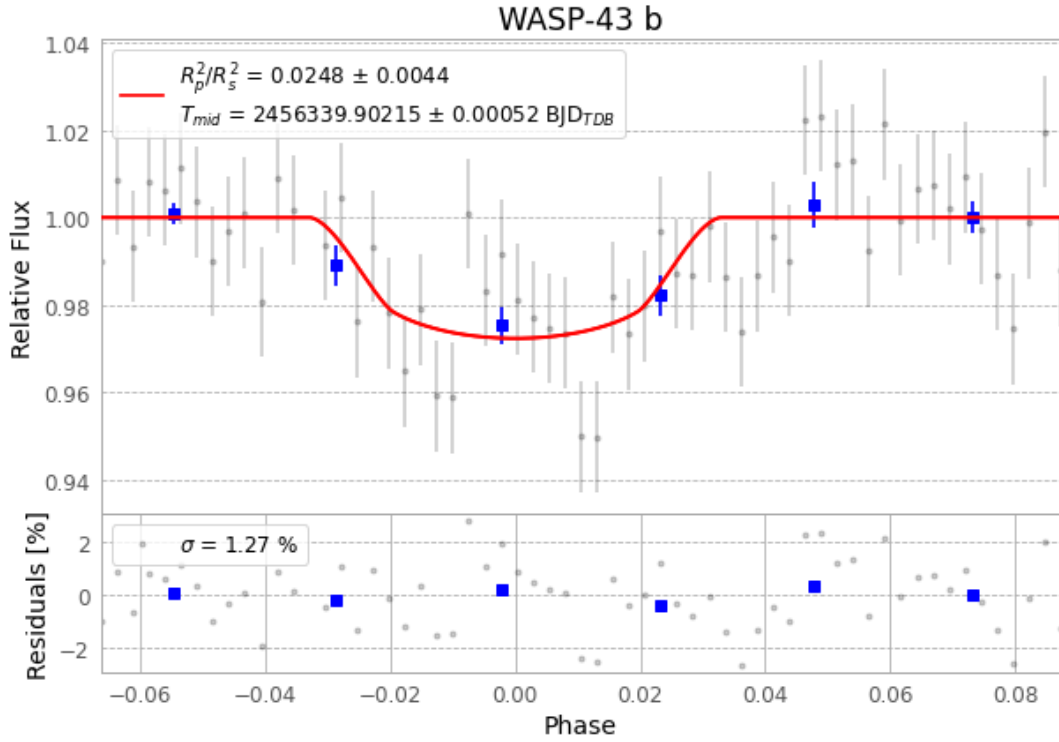


Figure 1 — FinalLightCurve_WASP-43 b_2013-02-16.png (SHA-256 67de52f4832b6ec1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 326], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 304a171567d1e0af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), WASP-43130216081212.FITS → WASP-43130216111212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130216.log (0f9f9c04ebce...), FinalLightCurve_WASP-43 b_2013-02-16.png (67de52f4832b...), FinalLightCurve_WASP-43 b_2013-02-16.csv (210d3e15a9c0...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:34Z.